

AP EAMCET-2019-Engineering-22 Apr-Shift-1
Questions Chemistry

121. If the kinetic energy of an electron of mass 9.0×10^{-31} kg is 2.0×10^{-25} J, the wavelength of the electron in nm is approximately

- (1) 1004.3
- (2) 1204.3
- (3) 1104.3
- (4) 994.3

122. If a suitable photon is employed to locate an electron (mass = 9.1×10^{-31} kg) in an atom within a distance of 10.98 nm, the uncertainty involved in the measurement of its velocity in ms^{-1} is

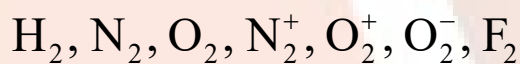
- (1) $\frac{1.6565 \times 10^{16}}{\pi}$
- (2) $\frac{1.6565 \times 10^4}{\pi}$
- (3) $\frac{1.6565 \times 10^{-8}}{\pi}$

$$(4) \frac{1.6565 \times 10^8}{\pi}$$

123. Which one of the following is the correct order of ionic radii?

- (1) $\text{Pr}^{3+} > \text{Gd}^{3+} > \text{Tm}^{3+}$
- (2) $\text{Pr}^{3+} < \text{Gd}^{3+} < \text{Tm}^{3+}$
- (3) $\text{Pr}^{3+} > \text{Tm}^{3+} > \text{Gd}^{3+}$
- (4) $\text{Pr}^{3+} < \text{Tm}^{3+} < \text{Gd}^{3+}$

124. Observe the following molecules/ions



Identify the correct statement

- (1) $\text{H}_2, \text{N}_2, \text{O}_2, \text{F}_2$ show diamagnetic property
- (2) $\text{O}_2, \text{O}_2^+, \text{O}_2^-, \text{N}_2^+$ show paramagnetic property
- (3) $\text{N}_2, \text{F}_2, \text{O}_2^+, \text{O}_2^-$ show diamagnetic property
- (4) $\text{H}_2, \text{N}_2^+, \text{O}_2^+, \text{O}_2^-$ show paramagnetic property

125. Match the following.

List-I	List-II
A) BrF_5	I) AB_4E , see-saw

B) SF_4	II) AB_4E_2 , square planar
C) XeF_4	III) AB_5E , square pyramid
D) ClF_3	IV) AB_3E_2 , T-shape
	V) AB_5E_2 , square pyramid

(1) $\frac{\underline{\text{A}}}{\underline{\text{V}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{IV}}}$

(2) $\frac{\underline{\text{A}}}{\underline{\text{III}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{V}}}$

(3) $\frac{\underline{\text{A}}}{\underline{\text{III}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{II}}}$ $\frac{\underline{\text{D}}}{\underline{\text{IV}}}$

(4) $\frac{\underline{\text{A}}}{\underline{\text{V}}}$ $\frac{\underline{\text{B}}}{\underline{\text{I}}}$ $\frac{\underline{\text{C}}}{\underline{\text{III}}}$ $\frac{\underline{\text{D}}}{\underline{\text{II}}}$

126. If the most probable speed of CO_2 at 27°C is 400 m s^{-1} , the root mean square velocity of CO_2 at the same temperature in m s^{-1} is approximately

(1) 600

(2) 490

(3) 267

(4) 245

127. 20 mL of Fe^{2+} solution of certain concentration has completely reacted with 20 mL of 0.01 M $\text{K}_2\text{Cr}_2\text{O}_7$ in an acid medium. If 20 mL of same Fe^{2+} solution has reacted completely with 20 mL of KMnO_4 solution in acid medium, the molarity of KMnO_4 solution is

- (1) 0.01 M
- (2) 0.12 M
- (3) 0.10 M
- (4) 0.012 M

128. If K_p for the conversion of oxygen to ozone at 400 K is 1.0×10^{-30} , its standard Gibbs energy change in kJ mol^{-1} is approximately

- (1) 229.8
- (2) 114.9
- (3) -229.8
- (4) -114.9

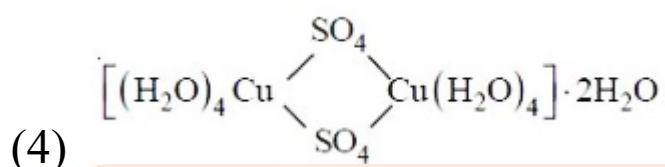
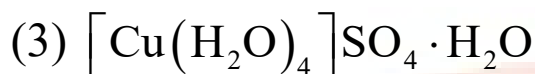
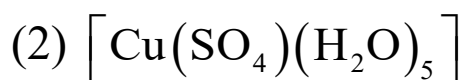
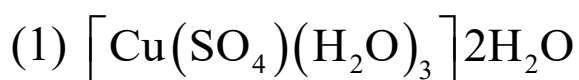
129. At 1000 K, the partial pressures of $\text{CO}_{2(g)}$ and $\text{CO}_{(g)}$ for the reaction $\text{CO}_{2(g)} + \text{C}_{(s)} \rightleftharpoons 2\text{CO}_{(g)}$ in a closed vessel at equilibrium are 0.15 and 0.60 bar respectively. The K_c for this reaction at the same temperature is approximately

- (1) 2.0×10^{-4}
- (2) 2.89×10^{-2}
- (3) 2.89×10^{-3}
- (4) 5.78×10^{-3}

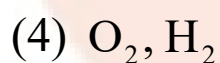
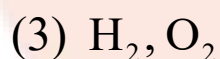
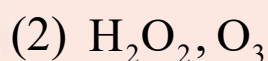
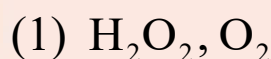
130. At T(K), if the ionization constant of ammonia in solution is 2.5×10^{-5} , the pH of 0.01 M ammonia solution and the ionization constant of its conjugate acid respectively at that temperature are ($\log 2 = 0.30$)

- (1) 10.7, 4.0×10^{-8}
- (2) 10.7, 4.0×10^{-10}
- (3) 3.3, 4.0×10^{-8}
- (4) 3.3, 4.0×10^{-8}

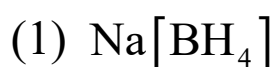
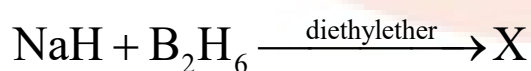
131. Which of the following correctly represents Copper sulphate pentahydrate?

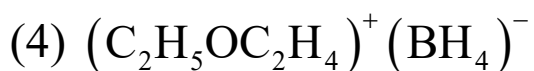
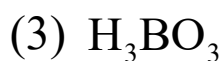


132. A compound (MO_2) of group I element (M) hydrolyses to form M^+ , OH^- , X and Y. When X reacts with I_2 in basic medium, the products formed are I^- , water and Y. Then X and Y are respectively



133. What is X in the following reaction?





134. Which one of the following is used as refrigerant for icecream and frozen foods?

(1) Dry CO

(2) Liquid CH_4

(3) Dry ice

(4) Liquid H_2

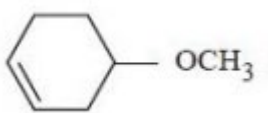
135. Acid rain is caused by the presence of X and Y in air. X, Y are respectively

(1) SO_2 , NO_2

(2) CFC, O_3

(3) CO, CFC

(4) SO_2 , O_3

136. IUPAC name of  is

(1) 5-methoxycyclohexene

(2) methoxycyclohex-3-ene

(3) methoxycyclohex-4-ene

(4) 4-methoxycyclohexene

137. Wurtz reaction of bromoethane gives n-butane. Sodium salt of X on heating with sodalime also results in n-butane. Compound X is

(1) $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

(2) $\text{CH}_3(\text{CH}_2)_3\text{COOH}$

(3) $\text{CH}_3(\text{CH}_2)_4\text{COOH}$

(4) $\text{CH}_3\text{CH}_2\text{COOH}$

138. Identify ortho and para directing groups from the following

–CHO

I

–NHCOCH₃

II

–OCH₃

III

–SO₃H

IV

(1) III, IV

(2) II, III

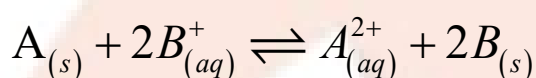
(3) II, IV

(4) I, IV

139. If 0.5 mol of a metal forms hexagonal close packed structure, the total number of voids, and tetrahedral voids respectively in mol are
- (1) 1.5, 1.0
 - (2) 1.0, 0.5
 - (3) 1.0, 1.5
 - (4) 0.5, 1.0
140. At 300 K an ideal solution is formed by mixing 460 g of toluene with 390 g benzene. If the vapour pressure of pure toluene and benzene at 300 K are 32 and 40 mm respectively, the mole fraction of toluene in vapour phase is
- (1) 0.196
 - (2) 0.588
 - (3) 0.294
 - (4) 0.444
141. If the aqueous solution contains 9% and 1% (W/W) of two non-volatile, non electrolytes X (molecular weight 180) and Y (molecular weight 50) respectively, the boiling point of solution in °C approximately is ($K_b = 0.52 \text{ K kg mol}^{-1}$)

- (1) 101.4
- (2) 100.4
- (3) 102.4
- (4) 100.8

142. If the E_{cell}^0 of an equilibrium reaction



at 298 K is 0.59 V, the equilibrium constant K_c is

- (1) 1.0×10^{10}
- (2) 1.0×10^2
- (3) 1.0×10^{-20}
- (4) 1.0×10^{20}

143. Which one of the following statements is correct?

- (1) In collision theory $e^{-E_a/RT}$ corresponds to the fraction of molecules that have energy equal; or greater than E_a
- (2) The number of collisions of reacting molecules per second per unit volume of the reaction mixture is activated complex
- (3) Molecularity is the number of molecules involved in a complex reaction
- (4) A catalyst catalyses the non-spontaneous reaction

144. Which one of the following enzymes convert proteins into aminoacids?

- (1) Maltase
- (2) Pepsin
- (3) Trypsin
- (4) Zymase

145. X_2 is used in the refining of Ti metal by van-Arkele method.

Y_2 does not liberate O_2 from water and does not form HY and HOY with water. X_2 and Y_2 are respectively

- (1) I_2, Cl_2
- (2) Cl_2, I_2
- (3) I_2, I_2
- (4) Cl_2, Cl_2

146. Which one of the following statements is **not** correct regarding phosphine?

- (1) It is a weak base
- (2) It reacts with $CuSO_4$ solution to form $CuHPO_4$
- (3) It is formed by the reaction of Ca_3P_2 with HCl
- (4) it is used in smoke screens.

147. Aqueous sulphite reacts with dilute sulphuric acid to form $X(g)$. The liberated $X(g)$ is passed into acidified $KMnO_4$ solution. What is the oxidation state of Mn in the product formed?

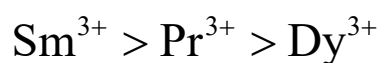
- (1) +6
- (2) +4
- (3) +2
- (4) +3

148. Phosphorus reacts with SO_2Cl_2 to form PCl_5 and X. At 723 K, HCl gas reacts with O_2 in the presence of $CuCl_2$ to form water and Y. Y reacts with X in water to form two acids A and B. What are A and B respectively?

- (1) H_2SO_4 , HCl
- (2) H_2SO_4 , HNO_3
- (3) HCl , HNO_3
- (4) H_2SO_4 , H_3PO_4

149. Identify the correct statements from the following

i. The ionic radius of Pr^{3+} , Dy^{3+} and Sm^{3+} follow the order,



ii. Eu^{2+} acts as strong reducing reagent

iii. Pu exhibits +7 oxidation state

(1) i, ii only

(2) i, iii only

(3) i, ii, iii

(4) ii, iii only

150. If the crystal field splitting energy of a tetrahedral complex (Δ_t) of the type $[\text{ML}_4]^{n+}$ is x eV, what is the crystal field splitting energy with respect to an octahedral complex, $[\text{ML}_6]^{n+}$?

(1) $\frac{9x}{4}$ eV

(2) $\frac{9x}{8}$ eV

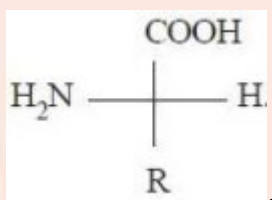
(3) $\frac{4x}{9}$ eV

(4) $\frac{4x}{5}$ eV

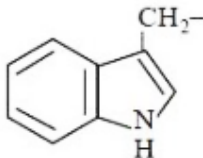
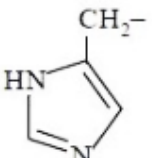
151. The monomers of nylon 6, 6 (X) and terylene (Y) are

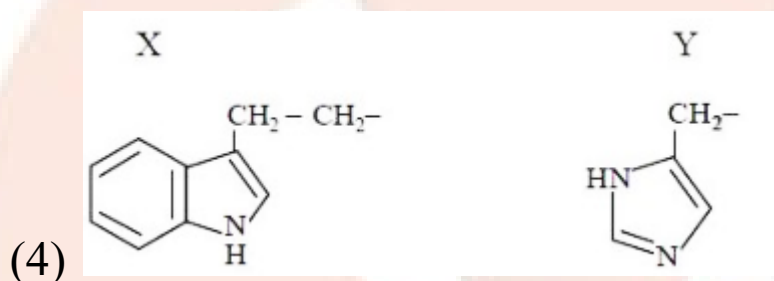
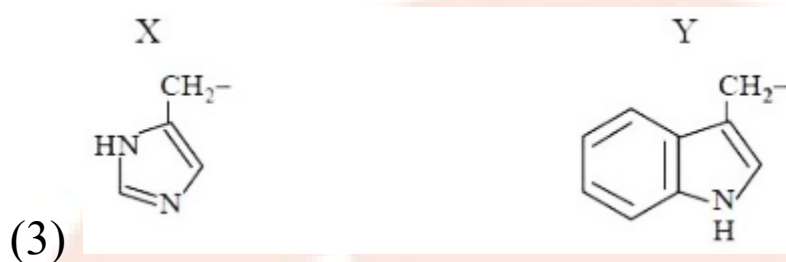
- (1) $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (X), $\text{HO}(\text{CH}_2)_6\text{COOH}$ (X), $\text{HO}(\text{CH}_2)_2\text{OH}$ (Y), $\text{HO}-\text{C}_6\text{H}_4-\text{COOH}$ (Y)
- (2) $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (X), $\text{HO}(\text{CH}_2)_4\text{COOH}$ (X), $\text{HO}(\text{CH}_2)_2\text{OH}$ (Y), $\text{HO}-\text{C}_6\text{H}_3(\text{COOH})_2-\text{COOH}$ (Y)
- (3) $\text{H}_2\text{N}(\text{CH}_2)_4\text{COOH}$ (X), $\text{H}_2\text{N}(\text{CH}_2)_6\text{COOH}$ (X), $\text{HO}(\text{CH}_2)_2\text{OH}$ (Y), $\text{HO}-\text{C}_6\text{H}_4-\text{COOH}$ (Y)
- (4) $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ (X), $\text{HO}(\text{CH}_2)_4\text{COOH}$ (X), $\text{HO}(\text{CH}_2)_2\text{OH}$ (Y), $\text{HOOC}-\text{C}_6\text{H}_4-\text{COOH}$ (Y)

152. The general structural formula of α -Amino acid is



The group R in tryptophan (X) and histidine (Y) are respectively

- (1) X :  Y : 



153. Which of the following statements are correct?

- i. Binding of inhibitor at allosteric site changes the shape of the active site
 - ii Shape of the receptor does not changes after attachment of chemical messenger to it
 - iii A chemical messenger gives message to the cell by entering it
 - iv Erthyromycin is an example of bacteriostatic antibiotic
- (1) i, ii
- (2) ii, iii

(3) i, iv

(4) iii, iv

154. Assertion (A) : S_N1 hydrolysis of optically active 2-bromooctane results in the formation of $(\pm)$ -octan-2-ol

Reason (R) : The reaction proceeds through a planar carbocation which can be attacked by the nucleophile from either side

The correct answer is

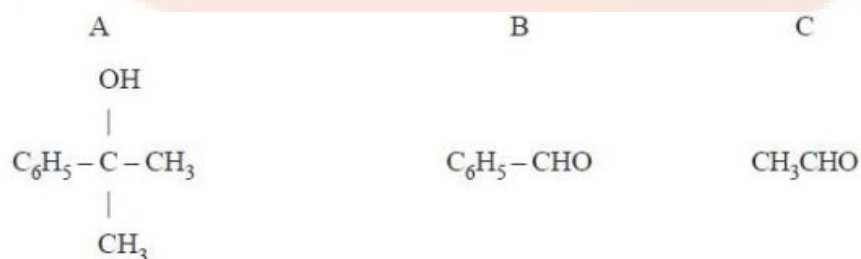
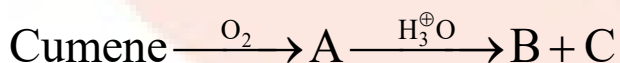
(1) (A) and (R) are correct, (R) is the correct explanation of (A)

(2) (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) (A) is correct but (R) is not correct

(4) (A) is not correct but (R) is correct

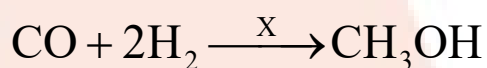
155. What are A, B and C in the following reactions?



(1)

- (2)
- | | | |
|---|--|----------------------|
| A | B | C |
| $\text{C}_6\text{H}_5-\text{CH}-\text{O}-\text{OH}$ | $\text{C}_6\text{H}_5-\text{CH}-\text{OH}$ | H_2O |
| | | |
| CH_3 | CH_3 | |
- (3)
- | | | |
|--|----------------------------------|----------------------------|
| A | B | C |
| CH_3 | | |
| | | |
| $\text{C}_6\text{H}_5-\text{C}-\text{O}-\text{OH}$ | $\text{C}_6\text{H}_5-\text{OH}$ | $(\text{CH}_3)_2\text{CO}$ |
| | | |
| CH_3 | | |
- (4)
- | | | |
|---|---|----------------------|
| A | B | C |
| CH_3 | CH_3 | |
| | | |
| $\text{C}_6\text{H}_5-\text{CH}-\text{CH}_2-\text{O}-\text{OH}$ | $\text{C}_6\text{H}_5-\text{CH}-\text{CH}_2\text{OH}$ | H_2O |

156. What is X in the following reaction?



- (1) 623 K/300 atm
 (2) $\text{KMnO}_4/\text{H}^\oplus$
 (3) Zn/Δ
 (4) $\text{ZnO}-\text{Cr}_2\text{O}_3$, 200-300 atm, 573-673 K

157. Alkenes (X) and carbonyl compounds (Y) participate in which of the following addition reactions?

- | | |
|-------------------|---------------|
| X | Y |
| (1) Electrophilic | Electrophilic |

X	Y
(2) Nucleophilic	Electrophilic

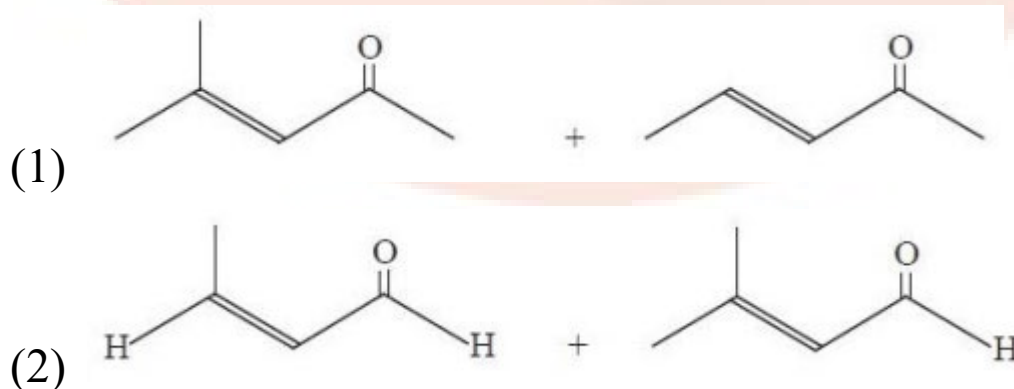
X	Y
(3) Electrophilic	Nucleophilic

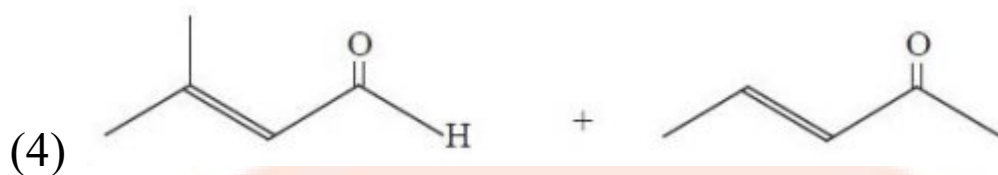
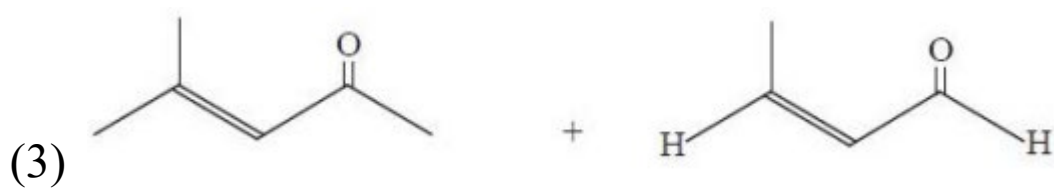
X	Y
(4) Nucleophilic	Nucleophilic

158. Which one of the following is used for purification of aldehydes?

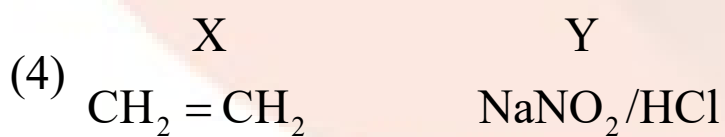
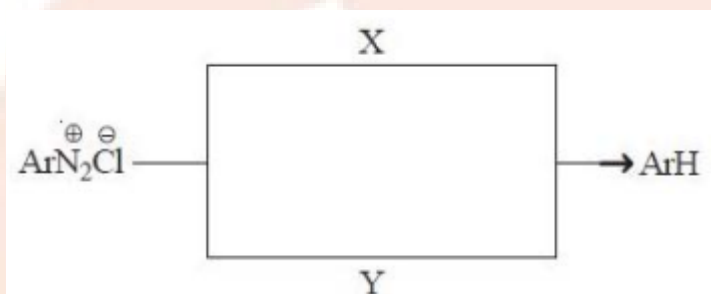
- (1) NaOCl
- (2) NaHSO_3
- (3) $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$
- (4) Na_2SO_4

159. Mixed aldol products obtained from aldol condensation of ethanal and propanone are





160. What are X and Y in the following reactions





AP EAMCET-2019-Engineering-22 Apr-Shift-1
Solutions – Chemistry

121. The wavelength is calculated as:

$$\begin{aligned}\lambda &= \frac{h}{\sqrt{2m(KE)}} \\ &= \frac{6.63 \times 10^{-34} \text{ Js}}{\sqrt{2 \times 9.0 \times 10^{-31} \text{ kg} \times 2 \times 10^{-25} \text{ J}}} \\ &= 11.043 \times 10^{-6} \text{ m}\end{aligned}$$

The conversion of wavelength into nm is shown below.

$$\begin{aligned}\lambda &= \frac{11.043 \times 10^{-6}}{10^{-9}} \text{ nm} \\ &= 1104.3 \text{ nm}\end{aligned}$$

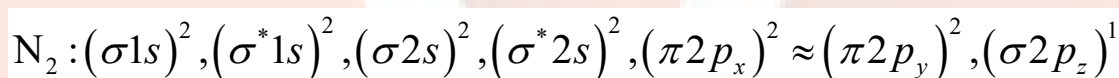
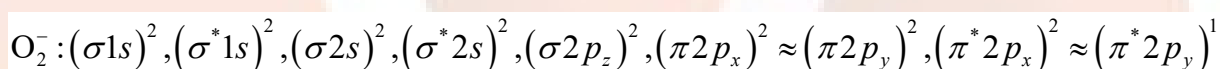
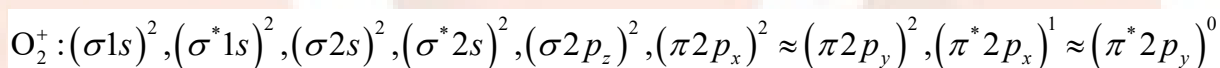
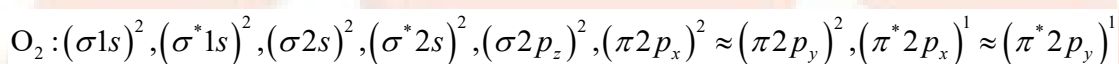
122. The uncertainty in velocity can be derived by using Heisenberg uncertainty principle as shown below.

$$\begin{aligned}\Delta x \cdot \Delta v &= \frac{h}{4\pi m} \\ &= \frac{6.634 \times 10^{-34} \text{ Js}}{4 \times \pi \times 9.1 \times 10^{-31} \text{ kg} \times 1.098 \times 10^{-9} \text{ m}} \\ &= \frac{1.6565 \times 10^4}{\pi}\end{aligned}$$

123. In 4-*f* series, due to the lanthanide contraction, the atomic number goes on increasing from La to Lu. As a result, atomic and ionic radii decreases. Therefore, the correct order is:

$$\text{Pr}^{3+} > \text{Gd}^{3+} > \text{Tm}^{3+}$$

124. The electronic configuration of the species which are having paramagnetic property (unpaired electrons) are shown below:



125. (A) The empirical formula of BrF_5 is AB_5E . Its hybridization is sp^3d^2 and shape is square pyramidal.

(B) The empirical formula of SF_4 is AB_4E . Its hybridization is sp^3d and shape is see saw.

(C) The empirical formula of XeF_4 is AB_4E_2 . Its hybridization is sp^3d^2 and shape is square planar.

(D) The empirical formula of ClF_3 is AB_3E_2 . Its hybridization is sp^3d and shape is T-shape.

Therefore,

A-III, B-I, C-II, D-IV

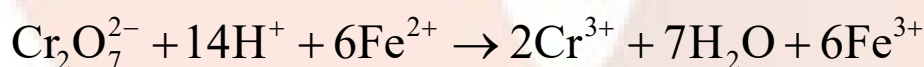
126. We can calculate the root mean square velocity by the ratio of:

$$v_{\text{rms}} : v_{\text{avg}} : v_{\text{most probable}}$$

$$1 : 0.92 : 0.82$$

$$\begin{aligned} v_{\text{rms}} &= \frac{v_{\text{most probable}}}{0.82} \\ &= \frac{400}{0.82} \\ &= 487.8 \\ &\approx 490 \text{ ms}^{-1} \end{aligned}$$

127. The reaction taking place in acidic medium between Fe^{2+} and $\text{Cr}_2\text{O}_7^{2-}$ is shown below.



The reaction taking place in acidic medium between Fe^{2+} and MnO_4^- is shown below.



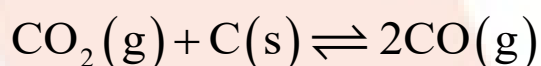
By applying the dilution formula:

$$\begin{aligned}M_1 \times V_1 \times n_1 &= M_2 \times V_2 \times n_2 \\0.01 \times 20 \times 6 &= M_2 \times 20 \times 5 \\M_2 &= \frac{0.01 \times 6}{5} \\&= 0.012 \text{ M}\end{aligned}$$

128. The standard Gibbs energy change is calculated as:

$$\begin{aligned}\Delta G^\circ &= -RT \ln K_p \\&= -0.008314 \text{ kJ/mol} \cdot \text{K} \times 400 \text{ K} \times \ln(1.0 \times 10^{-30}) \\&= -3.3256 \text{ kJ/mol} \times -69.0775 \\&= 229.724 \text{ kJ/mol} \\&\approx 229.8 \text{ kJ/mol}\end{aligned}$$

129. The given reaction is:



$$\begin{aligned}\Delta n &= \text{Moles of gaseous products} - \text{Moles of gaseous reactants} \\&= 2 - 1 \\&= 1\end{aligned}$$

First we need to calculate K_p as shown below.

$$\begin{aligned}
 K_p &= \frac{(p_{CO})^2}{(p_{CO_2})} \\
 &= \frac{(0.6)^2}{0.15} \\
 &= 2.4
 \end{aligned}$$

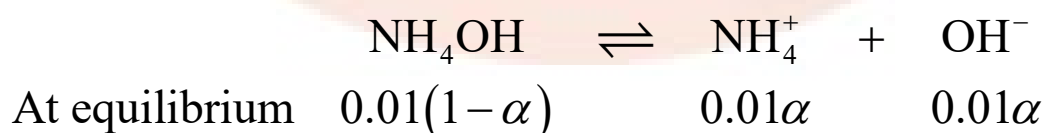
Now, the value of K_c is calculated as:

$$\begin{aligned}
 K_p &= K_c (RT)^{\Delta n} \\
 2.4 &= K_c (0.0821 \times 1000)^1 \\
 K_c &= \frac{2.4}{0.0821 \times 1000} \\
 &= 2.89 \times 10^{-2}
 \end{aligned}$$

130. The ionization constant of conjugate acid is calculated as:

$$\begin{aligned}
 K_a(NH_4^+) \cdot K_b(NH_3) &= 10^{-14} \\
 K_a(NH_4^+) &= \frac{10^{-14}}{K_b(NH_3)} = \frac{10^{-14}}{2.5 \times 10^{-5}} \\
 &= 4 \times 10^{-10}
 \end{aligned}$$

The dissociation reaction of ammonia is as follows:



The degree of ionization is:

$$\begin{aligned}\alpha &= \sqrt{\frac{K_b}{C}} \\ &= \sqrt{\frac{2.5 \times 10^{-5}}{0.01}} \\ &= 0.05\end{aligned}$$

Therefore concentration of hydroxide ion becomes:

$$\begin{aligned}[\text{OH}^-] &= C\alpha \\ &= 0.01 \times 0.05 \\ &= 0.0005\end{aligned}$$

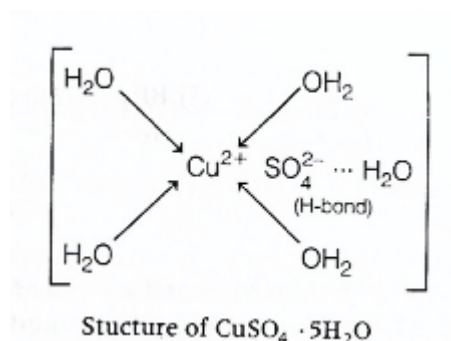
The pH of the ammonia solution is calculated as:

$$\begin{aligned}\text{pH} &= 14 - \text{pOH} \\ &= 14 - (-\log[\text{OH}^-]) \\ &= 14 + \log(0.005) \\ &= 14 + (-3.3) \\ &= 10.7\end{aligned}$$

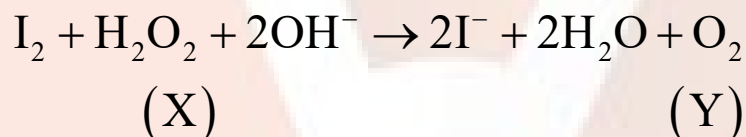
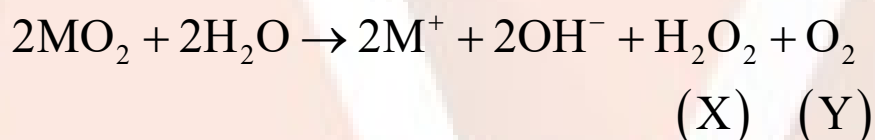
Therefore, pH is 10.7 and ionization constant is equal to 4×10^{-10}

131. Copper sulphate pentahydrate is represented by the molecular formula of $[\text{Cu}(\text{H}_2\text{O})_4]\text{SO}_4 \cdot \text{H}_2\text{O}$. It has five molecules of water of crystallization. Four molecules of water are coordinated

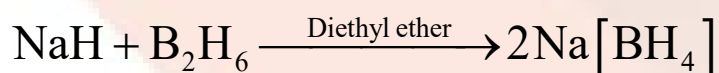
with Cu^{2+} and remaining one molecule of water coordinates through hydrogen bonding as shown below:



132. The reactions taking place are shown below.



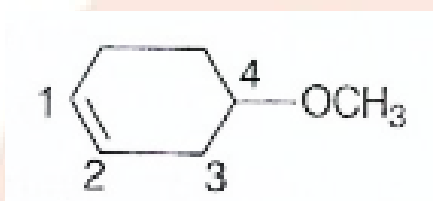
133. When sodium hydride undergoes a reaction with diborane, it forms sodium tetrahydroxoborate (X) as shown below.



134. Dry ice is used as refrigerant for ice cream and frozen foods. It preserves food items by using non cyclic refrigeration.

135. The presence of SO_2 and NO_2 in the atmosphere causes acid rain. These substances react with water, oxygen and other harmful chemicals to produce acid rain.

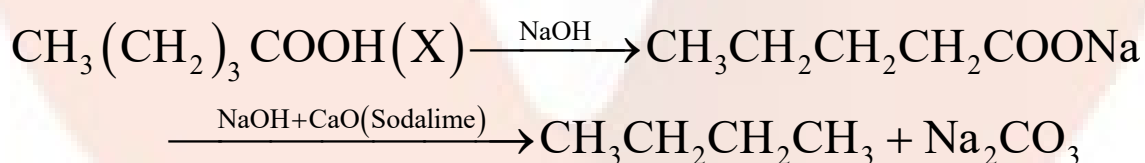
136. The IUPAC name of the given structure is 4-methoxycyclohexene as shown below.



137. The Wurtz reaction is shown below.



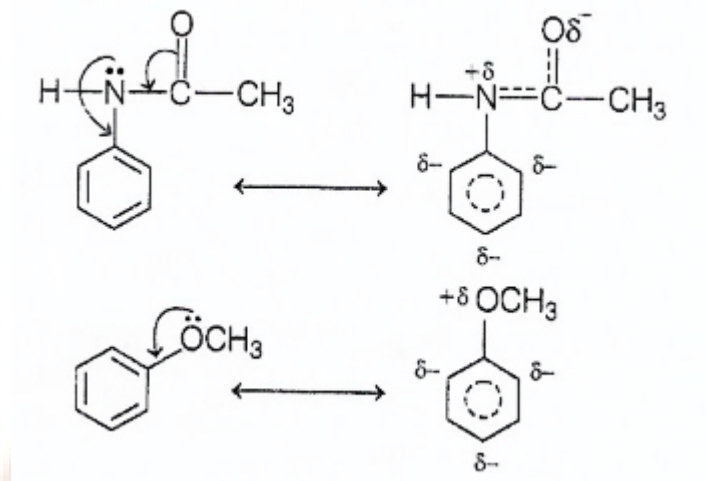
When sodium salt of pentanoic acid (X) reacts with sodalime and heated, it forms n-butane as shown below.



Therefore, X is $\text{CH}_3(\text{CH}_2)_3\text{COOH}$

138. The ortho and para directing groups are $-\text{NHCOCH}_3$ and $-\text{OCH}_3$. These groups are known as electron donating groups as these groups increases the density of

electrons especially at ortho and para positions of the aromatic ring as follows:



139. The number of atoms is calculated as:

$$\begin{aligned}\text{For } 0.5 \text{ mol} &= 0.5 \times 6.022 \times 10^{23} \\ &= 3.011 \times 10^{23}\end{aligned}$$

Tetrahedral voids is equal to $2 \times$ Number of atoms of metal

Octahedral voids is equal to number of atoms of metal

Therefore,

$$\begin{aligned}\text{Number of tetrahedral voids} &= 2 \times 3.011 \times 10^{23} \\ &= 6 \times 10^{23} \\ &= 1 \text{ mol}\end{aligned}$$

$$\begin{aligned}\text{Number of octahedral voids} &= 3.011 \times 10^{23} \\ &= 0.5 \text{ mol}\end{aligned}$$

The total number of voids is $1 + 0.5 = 1.5 \text{ mol}$

140. The mole fraction of toluene (A) is calculated as:

$$\begin{aligned}x_A &= \frac{\frac{w_A}{M_A}}{\frac{w_A}{M_A} + \frac{w_B}{M_B}} \\&= \frac{\frac{460}{92}}{\frac{460}{92} + \frac{390}{78}} \\&= 0.5\end{aligned}$$

The mole fraction of benzene (B) is calculated as:

$$\begin{aligned}x_B &= 1 - x_A \\&= 1 - 0.5 \\&= 0.5\end{aligned}$$

The total pressure becomes equal to

$$\begin{aligned}p_{\text{total}} &= p_A^{\circ}x_A + p_B^{\circ}x_B \\&= (32 \times 0.5) + (40 \times 0.5) \\&= 36\end{aligned}$$

Mole fraction of toluene in vapour pressure (Y_T) is derived as:

$$p_T^{\circ} x_T = Y_T \cdot p_{\text{total}}$$

$$\begin{aligned} Y_T &= \frac{p_T^{\circ} x_T}{p_{\text{total}}} \\ &= \frac{32 \times \frac{1}{2}}{36} \\ &= 0.444 \end{aligned}$$

141. The change in boiling point is calculated as shown below.

$$\begin{aligned} \Delta T_b &= K_b [M_X + M_Y] \\ &= K_b \left[\frac{w_X}{M_X} + \frac{w_Y}{M_Y} \right] \times \frac{1000}{100} \\ &= 0.52 \left[\frac{9}{180} + \frac{1}{50} \right] 10 \\ &= 0.36 \end{aligned}$$

The boiling point of solution will become as:

$$\begin{aligned}T_b &= T_b^\circ + \Delta T_b \\&= 100 + 3.6 \\&= 100.36 \\&\approx 100.4\end{aligned}$$

142. The equilibrium constant is calculated as:

$$\begin{aligned}E_{\text{cell}}^\circ &= \frac{0.059}{n} \ln K \\0.59 &= \frac{2 \times 0.59}{0.059} \ln K \\\ln K &= 20 \\K &= 1 \times 10^{20}\end{aligned}$$

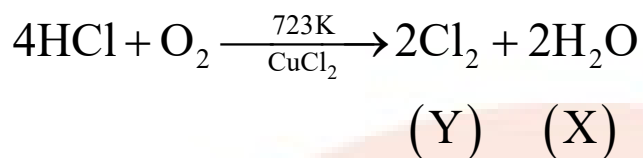
143. In collision theory, the term $e^{-E_a/RT}$ gives information about the fraction of molecules that are having energy equal to or greater than activation energy.

Therefore statement (1) is correct.

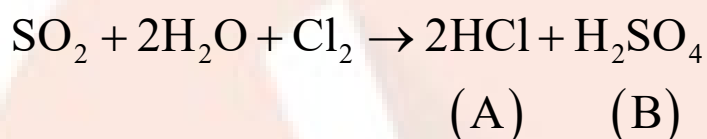
144. The enzyme trypsin changes proteins into amino acids. It activates the pancreatic enzymes. First, it convert proteins into dipeptides and then finally into amino acids.

145. The reactions are shown below.

When HCl gas undergoes a reaction with oxygen in the availability of CuCl_2 , it produces water and chlorine gas.



Now, when Y reacts with X in water, it produces HCl (A) and H_2SO_4 (B)



149. (i) The correct order of ionic radii is shown below.

$$\text{Pr}^{3+} > \text{Sm}^{3+} > \text{Dy}^{3+}$$

$$1.013 \text{ \AA} \quad 0.964 \text{ \AA} \quad 0.908 \text{ \AA}$$

(ii) Eu^{2+} acts as strong reducing agent as +2 oxidation state oxidizes to +3 easily.

(iii) Pu shows +7 oxidation state. It can use maximum eight electrons because its electronic configuration is $5f^6 6d^0 7s^2$.

Therefore, (ii) and (iii) statements are correct.

150. The expression between tetrahedral and octahedral crystal field splitting is shown below.

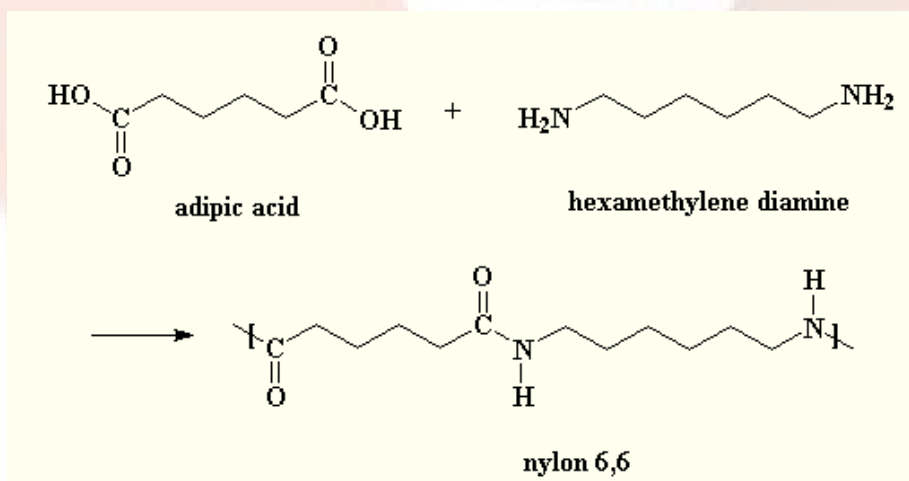
$$\Delta_t = \frac{4}{9} \Delta_o$$

$$\Delta_o = \frac{9}{4} \Delta_t$$

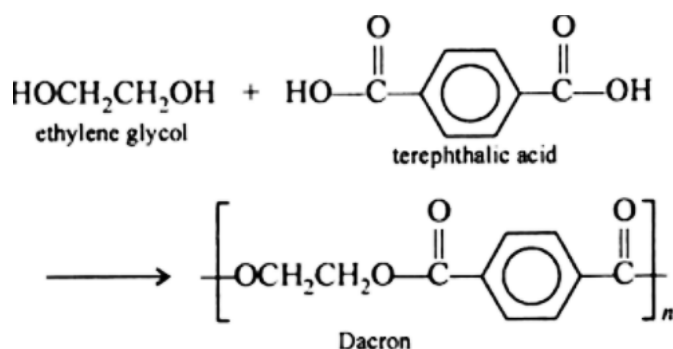
It is given that CFSE of tetrahedral complex of $[ML_4]^{n+}$ is x eV, therefore the CFSE of octahedral complex $[ML_6]^{n+}$ will become as:

$$\Delta_o = \frac{9x}{4}$$

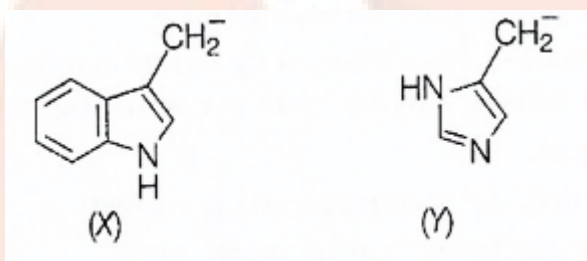
151. Nylon 6,6 is formed by the condensation reaction of hexamethylene diamine and adipic acid as shown below.



Dacron is formed by the condensation reaction of ethane-1,2-diol and terephthalic acid as shown below.

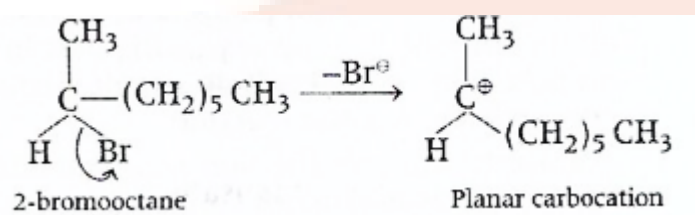


152. The group R in tryptophan (X) and histidine (Y) are shown below.



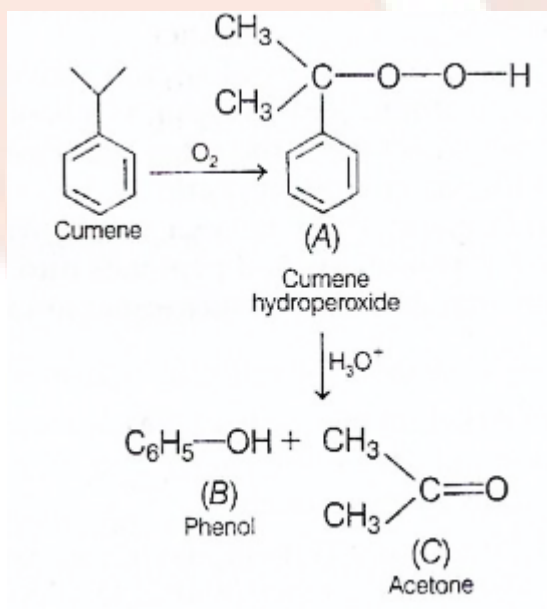
153. (i) The shape of the active sites is changed by the binding of inhibitor at allosteric site
- (ii) After attachment of chemical messenger to receptor, the shape of receptor gets changed.
- (iii) Without entering into the cell, a chemical messenger passes messages to the cell.
- (iv) Erythromycin is an example of bacteriostatic antibiotic.
- Therefore, statements (i) and (iv) are correct.

154. S_N1 hydrolysis of optically active 2-bromooctane forms $(\pm)$ -octan-2-ol as the reaction proceeds through a planar carbocation which can be attacked by the nucleophile from either side. The reaction is as follows:



Therefore, Assertion (A) and Reason (R) is the correct explanation of (A)

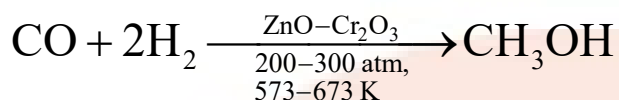
155. The products (A), (B) and (C) formed are shown below



156. When carbon monoxide reacts with hydrogen in the presence of reagents such as $\text{ZnO-Cr}_2\text{O}_3$ at a pressure of 200-300 atm and

at a temperature of 573-673 K, the formation of methanol takes place

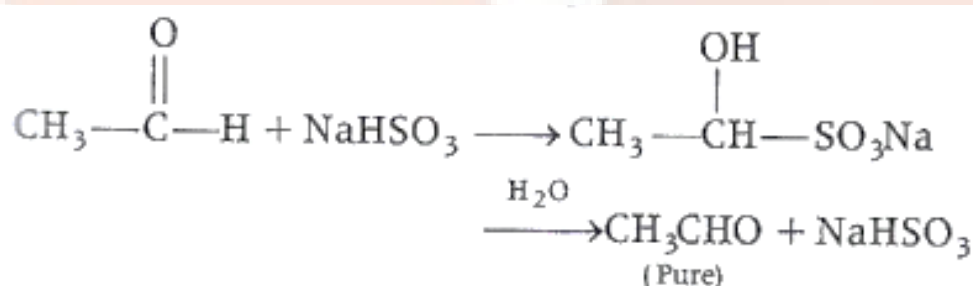
The reaction is shown below.



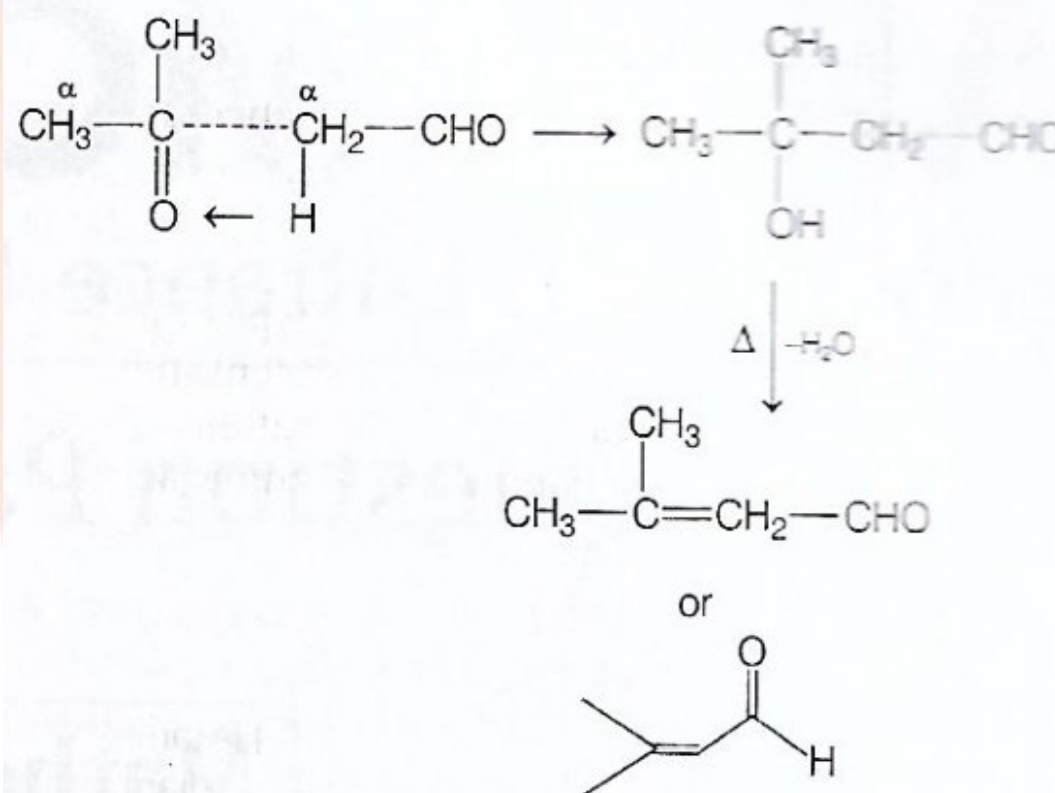
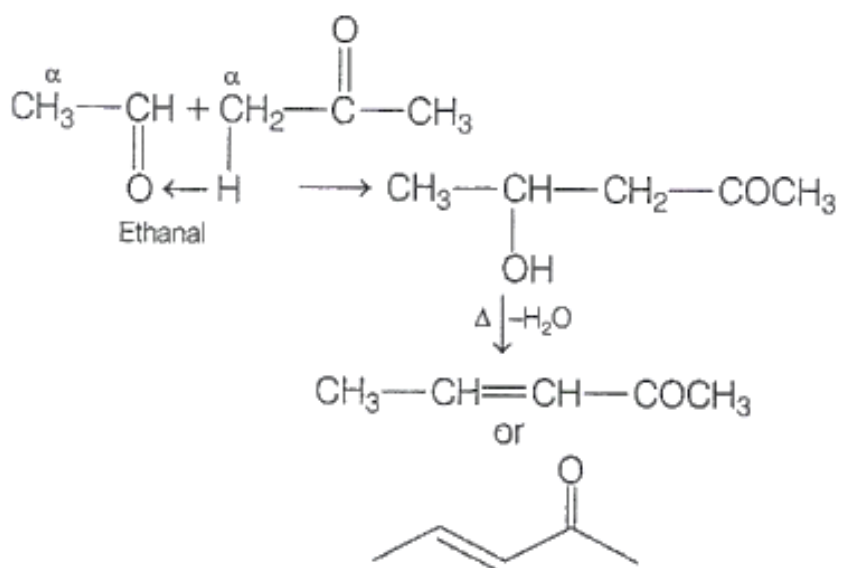
157. Alkenes (X) is electrophilic because alkenes contain π – electron cloud hence, they are highly reactive substances. Therefore, X shows electrophilic reactions.

Carbonyl compounds (Y) is nucleophilic because carbonyl compounds are highly polar in nature. Therefore, Y shows nucleophilic reactions.

158. NaHSO_3 is used for the purification of aldehydes. The reaction is shown below.



159. The aldol condensation reaction taking place between ethanal and propanone is as follows:



160. The products (x) and (Y) formed are shown below.

